

Logistic Regression: The Two Cultures

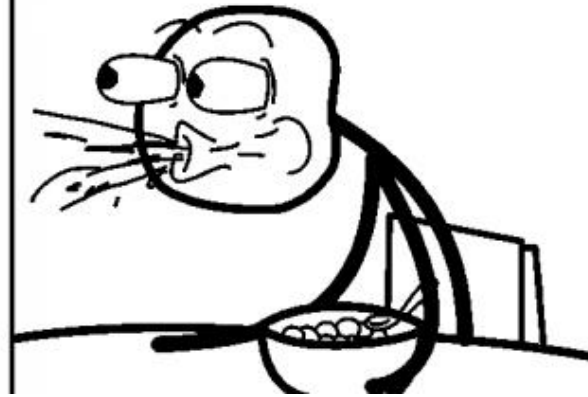
ML practitioner:

“Logistic regression
is a classification
algorithm!”



Statistician:

“ ... ”



Logistic Regression: The Two Cultures



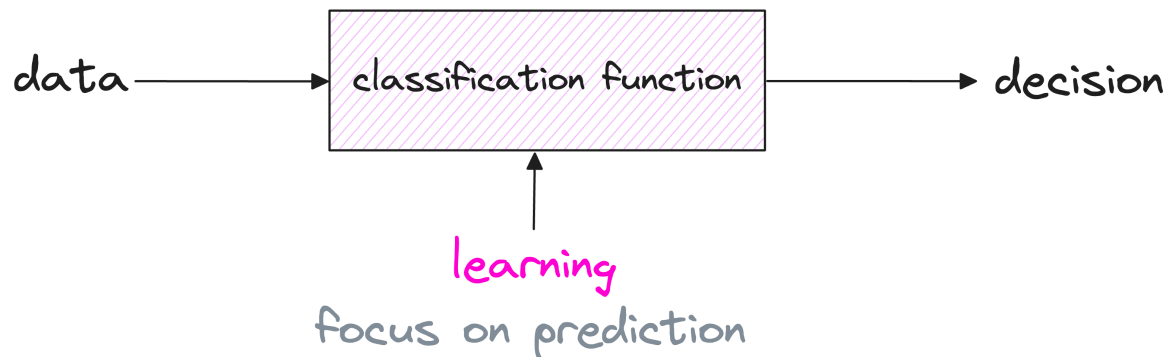
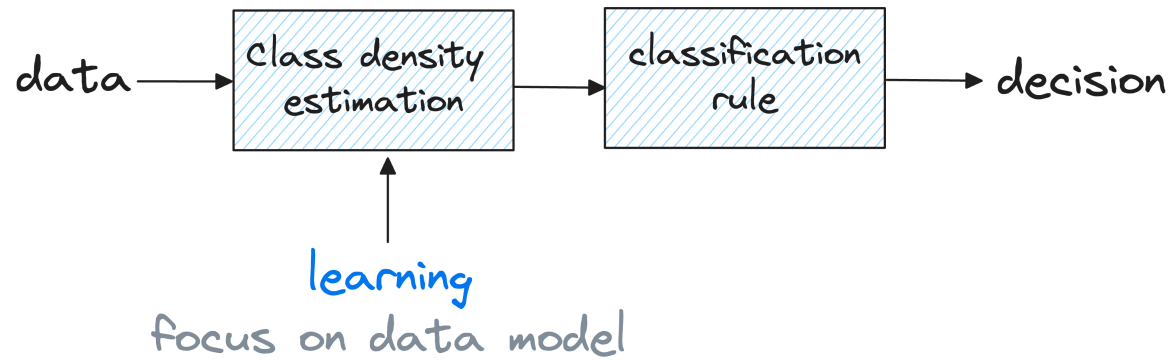
Is it correct to call a logistic regression classifier?

The answer to this question seems to go beyond a simple yes or no.

The classification vs. regression debate surrounding logistic regression is a topic of discussion among **ML practitioners** and **Statistics practitioners**. While logistic regression is commonly used for binary classification tasks, its underlying mechanics and interpretation do raise questions about its classification status.

- On the one hand, **ML practitioners** emphasize its conceptual similarity to other ML methods which help in dichotomous target variable prediction (e.g., SVM and other techniques which may not even produce probabilities)
- **Statistics practitioners**, on the other hand, highlight the uniqueness of this method and its entirely distinct characteristics that relegate the classification aspect to a secondary nature, if not rendering it entirely moot

Probabilistic vs Algorithmic Modeling Culture



- In his seminal paper¹, Leo Breiman, a renowned statistician and the creator of the Random Forest algorithm, highlighted the divergent approaches between traditional statistical modeling and machine learning, which is key to the Logistic Regression debate
- He notes that statisticians typically focus on developing parametric models that represent the data generation process (data model), while machine learning practitioners prioritize predictive accuracy

1. <https://users.csc.calpoly.edu/~dsun09/data401/readings/Breiman-Two-Cultures.pdf>

Statistical View on Logistic Regression

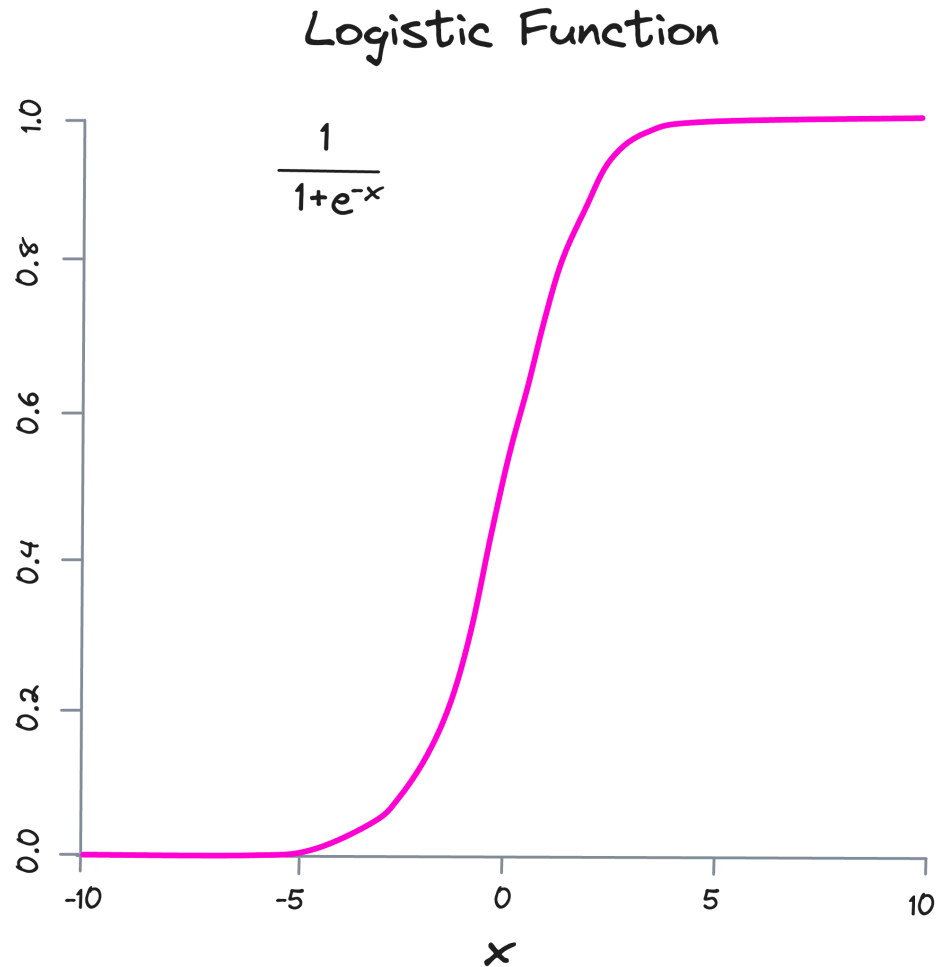


“The logistic model is a direct probability model”

- Logistic regression is a method most often used to study and identify explaining factors rather than to do individual prediction (the Two Cultures)
- Logistic regression **is not a classification algorithm on its own**. It is only a classification algorithm in combination with a decision rule (threshold) that makes the predicted probabilities of the outcome dichotomous (0 or 1)
- Whether engaging in credit risk scoring, weather forecasting, climate forecasting, marketing, diagnosis a patient's disease, or estimating a patient's prognosis, we do not want to use a classification method. Instead we want risk estimates with credible intervals or confidence intervals (Harrell 2017¹)
- Using classification tables for assessing goodness of fit is most appropriate when classification is a stated goal of the analysis (Hosmer and Lemeshow 2013²)

1. <https://www.fharrell.com/post/classification/>
2. <https://onlinelibrary.wiley.com/doi/book/10.1002/9781118548387>

Statistical View on Logistic Regression



- Logistic regression is an “old” statistical technique that has gained widespread adoption, particularly within the traditional statistical community. Its roots can be traced back to the work of Walker, Duncan, and Cox in the late 1950s.
- Unlike modern ML methods, which often prioritize predictive accuracy, Logistic Regression is primarily used for studying and identifying explanatory factors rather than individual predictions (The Data Modeling Culture).

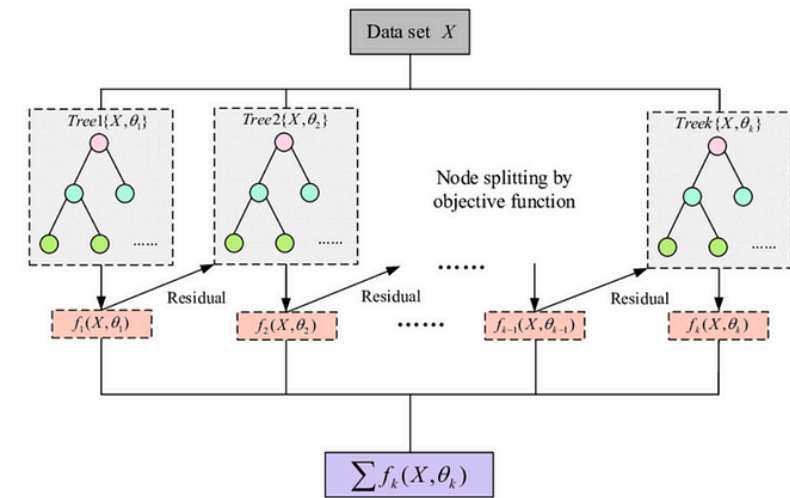
Evidence in Favor of Statistical View



Logistic Regression model use: Logistic regression is frequently utilized within risk models to estimate probabilities for events such as default or delinquency, rather than solely focusing on direct class predictions. This probabilistic interpretation aligns more closely with a regression-oriented approach

Log-odds modeling with regression: The logistic regression model estimates the log odds of the binary target variable. This suggests that it's more about capturing the relationship between independent variables and the likelihood of the target class rather than assigning a definitive class label

XGBoost Logistic Regression: Even within the gradient boosting framework, there is no classification. The modeling is handled as a regression problem. This is achieved by fitting residuals and producing output in the form of log-odds (margins)



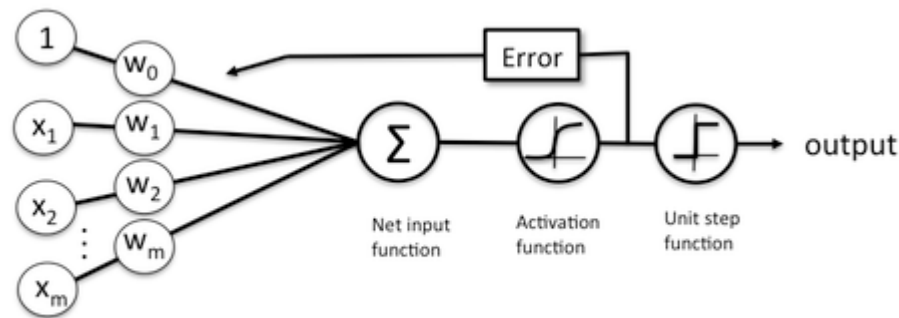
Flow chart of XGBoost

ML View on Logistic Regression

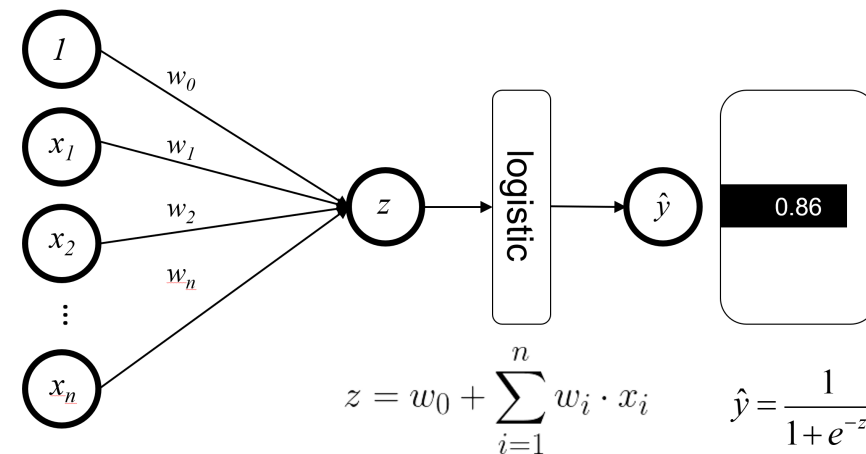


“Logistic regression is a probabilistic classifier”

- The chief value of logistic regression is that it provides an important theoretical window on the behavior of more complex classification methodologies (Friedman et al., 2000¹)
- The “classic” application of logistic regression model is binary classification (Raschka 2016²)



Schematic of a logistic regression classifier.



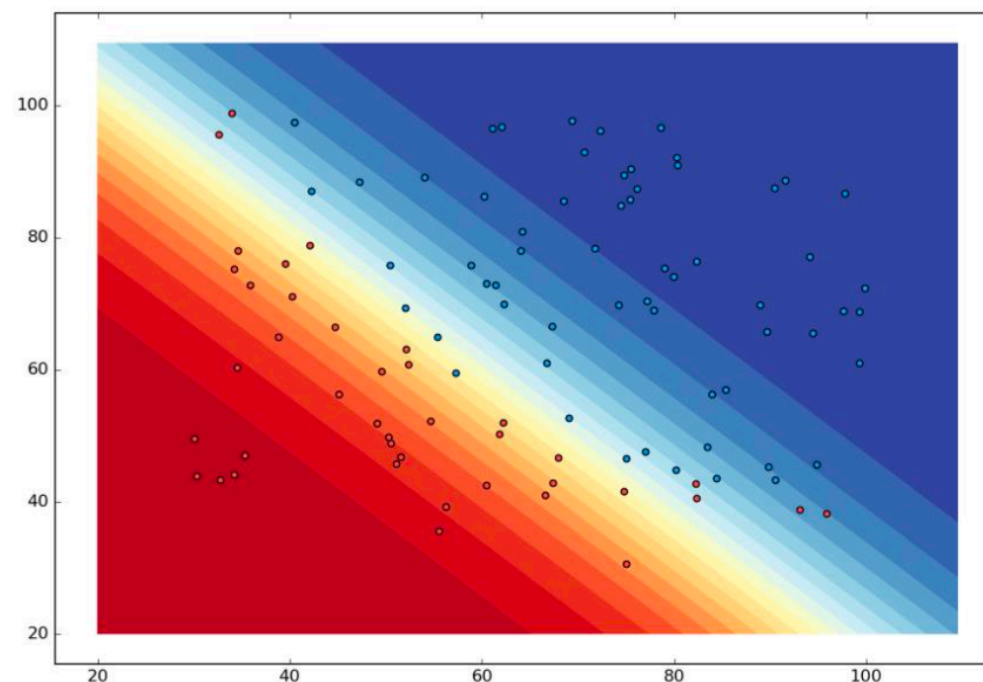
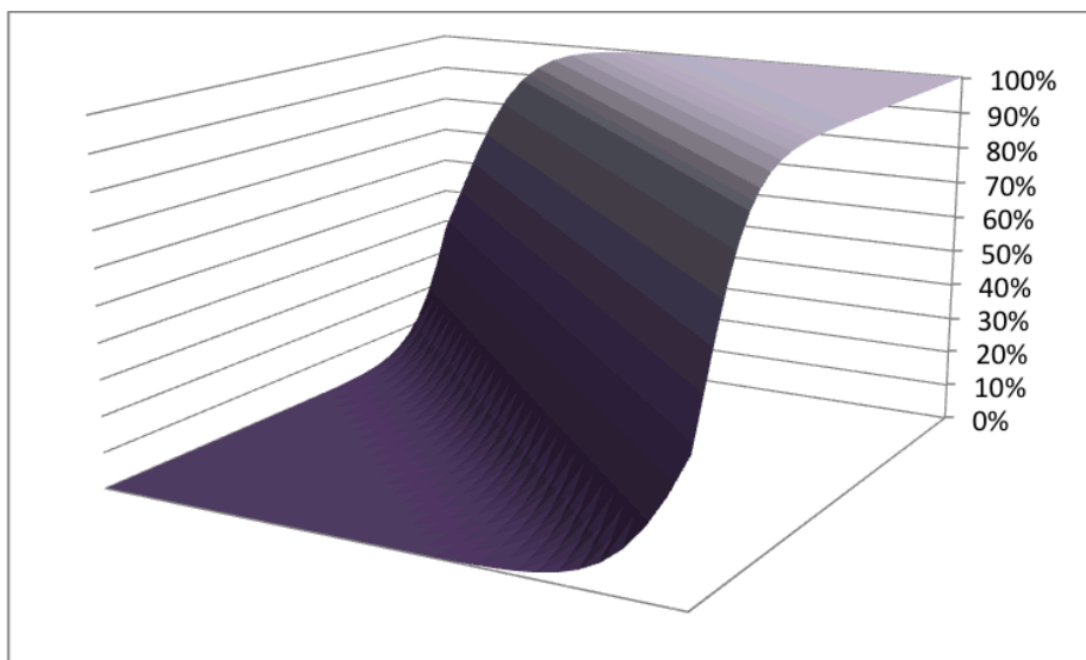
1. <https://hastie.su.domains/Papers/AdditiveLogisticRegression/alr.pdf>
2. <https://www.kdnuggets.com/2016/07/softmax-regression-related-logistic-regression.html>

ML View on Logistic Regression



Logistic Regression can be viewed as a model of a posterior class distribution for classification and can be interpreted as a probabilistic linear classifier. It is a fully probabilistic model, not only a “post-processing” of a linear classifier.

With Logistic Regression, we model the posterior distribution for classification in a two-class-setting by applying the logistic function to the linear classifier.



Evidence in Favor of ML View

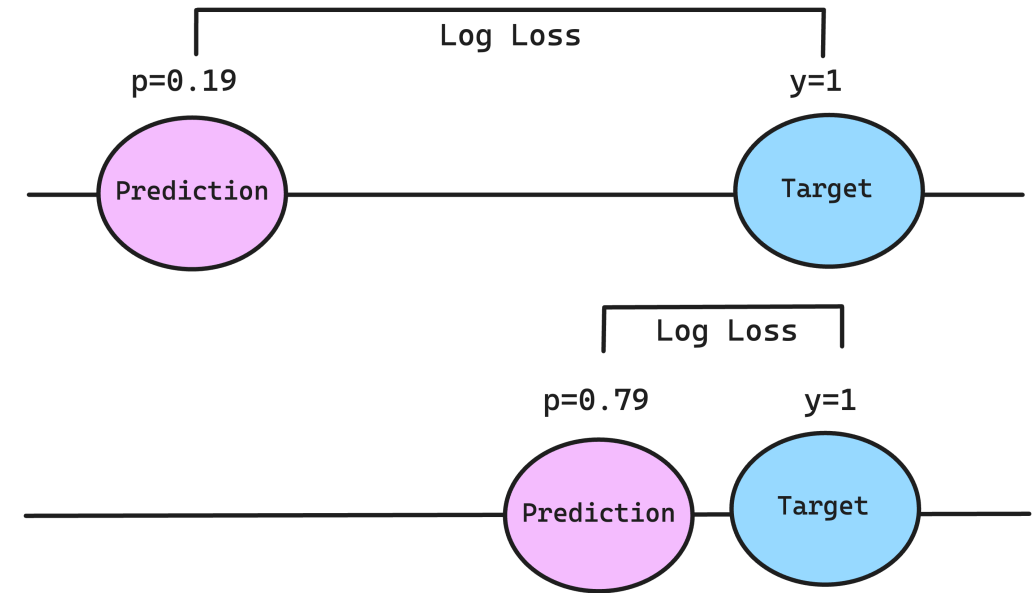


Maximum Likelihood Estimation (MLE), advanced by Ronald A. Fisher, is applicable to both classification (Linear Discriminant Analysis) and Logistic Regression. In case of Logistic Regression, we are finding the best weights (coefficients) that bring our predictions to observed labels as close as possible.

$$L(p) = \prod_{i=1}^n p_i^{y_i} \times (1-p_i)^{1-y_i}$$

$$LL(p) = \sum_{i=1}^n \log(p_i)^{y_i} + \log(1-p_i)^{1-y_i}$$

$$LL(p) = \sum_{i=1}^n y_i \times \log(p_i) + (1-y_i) \times \log(1-p_i)$$



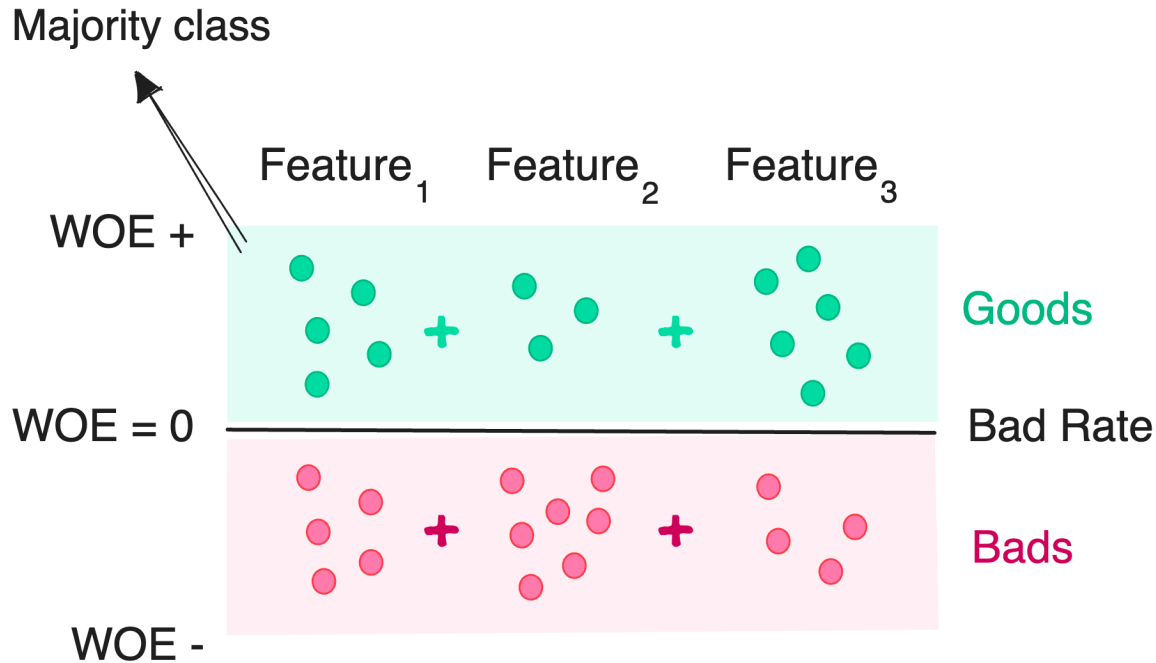
Log Loss is the objective function in Logistic Regression. It measures the difference between predicted probabilities and actual class labels. By minimizing Log Loss, we can improve the model's ability to accurately model probabilities of class membership.

Evidence in Favor of ML View



In the context of Weight-of-Evidence (WOE) in credit scoring, we do in fact try to “classify” instances into groups, and Information Value (IV) measures the class separability. This is similar to the Gini index in decision trees where based on the prevalence of class instances we classify observations within leaf nodes

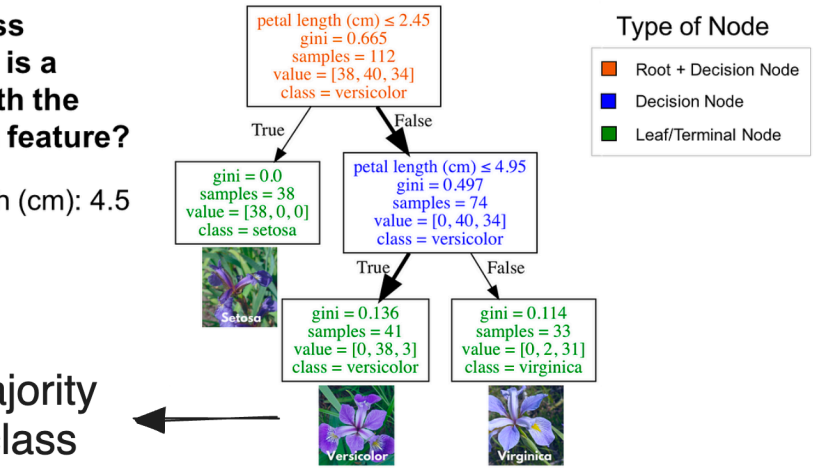
Classification example



What class (species) is a flower with the following feature?

petal length (cm): 4.5

Majority class



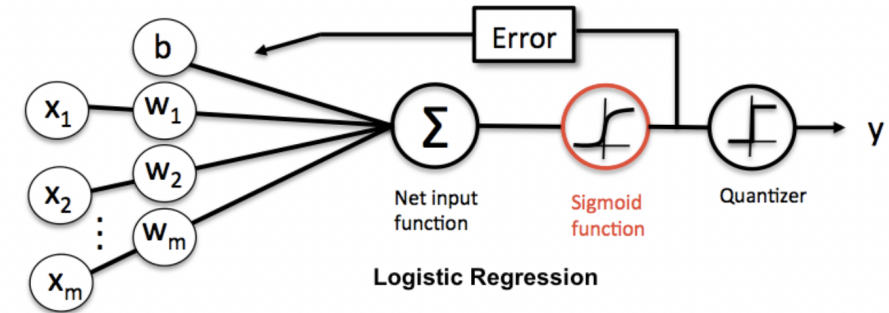
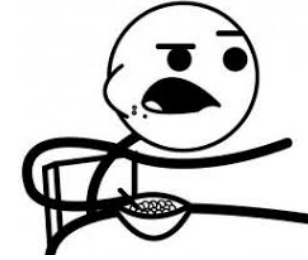
Species counts are: setosa=0, versicolor=38, virginica=3

Prediction is **versicolor** as it is the majority class

Bottom line



- While Logistic Regression is commonly employed for classification tasks, its operational mechanisms and real-world applications may challenge its classification status, triggering debates reminiscent of the 'Two Cultures' discourse introduced by L. Breiman
- A more balanced perspective could be to view Logistic Regression as a hybrid model that combines both classification and regression aspects: it models a binary event like other classifiers, but achieves it through different means (modelling of the posterior distribution vs. classification label)
- At its core, Logistic Regression operates as a direct probabilistic model. Rather than merely assigning class labels, it undertakes classification based on probability distributions, underlining its distinction from direct class label assignment in traditional classifiers.



Readings



Statistical View



ML View

